A Test for Instrumental Variable Validity Using a Partially Identified Correlation Restriction *

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Abstract

We develop a Correlation Restriction (CR) test to assess the validity of instrumental variables when the direction of endogeneity is known. Consistent with Gunsilius (2021), the test demonstrates that additional structural assumptions allow instrument validity to be assessed even in models with continuous endogenous variables. Extending DiTraglia and García-Jimeno (2021), our approach uses the joint correlation structure of the instrument, regressor, and outcome to construct confidence intervals for partially identified parameters via a modified union-bound procedure. Monte Carlo simulations and empirical applications, including analyses of returns to education, recidivism, and development, demonstrate that the CR test reliably detects invalid instruments and characterizes the range of exogeneity, enabling formal empirical evaluation of instrument validity.

Keywords: Endogeneity, instrumental variable validity, linear regression

JEL codes: C18, C26, C36, C52

1 Introduction

Instrumental variable (IV) estimation is a cornerstone of causal inference under endogeneity. Its validity rests on two conditions: (i) relevance, meaning that the instrument is correlated with the endogenous regressor, and (ii) exogeneity, requiring that it is uncorrelated with the structural error, $E[z'u] = 0$. While relevance is empirically verifiable, exogeneity is generally untestable because u is unobserved. Empirical practice therefore relies on institutional arguments and informal reasoning to justify instrument validity.

Pearl (1995) formalized this challenge, conjecturing that with continuously distributed endogenous variables, instrument validity is theoretically untestable. Subsequent work, including Gunsilius (2021), showed that introducing mild structural or independence assumptions can partially restore testability. Building on this insight, we demonstrate that imposing sign restrictions on the direction of endogeneity enables empirical testing of instrument validity in continuous settings. Embedding these restrictions within the data-implied correlation structure transforms an otherwise untestable condition into a falsifiable empirical restriction.

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Sign restrictions on $\text{corr}(x,u)$ are plausibly available in empirical settings where IV estimation is employed. Researchers often have domain-specific knowledge about the direction of endogeneity bias, even when its magnitude is unknown (Moon and Schorfheide, 2009). DiTraglia and García-Jimeno (2021) incorporate such sign and interval restrictions on regressor endogeneity as informative priors to narrow the identified set in their Bayesian framework. Similarly, Nevo and Rosen (2012) establish partial identification results by explicitly linking the sign of $\text{cov}(x,u)$ to structural parameters, thereby deriving bounds for causal effects when instruments may be invalid.

Our approach exploits the joint dependence structure linking the instrument, the endogenous regressor x, the outcome y, and the structural error u. Following DiTraglia and García-Jimeno (2021), these variables must satisfy

$$\rho_{zu} = \rho_{xz}\rho_{xu} - (\rho_{xy}\rho_{xz} - \rho_{yz})\sqrt{\frac{1-\rho_{xu}^2}{1-\rho_{xy}^2}},$$

where ρ_{ij} denotes the correlation between variables i and j. Given an admissible domain $\mathscr{D}$ for ρ_{xu} consistent with the known sign of $\text{corr}(x,u)$, this mapping defines an identified set $\mathscr{C} = g(\rho,\mathscr{D})$ for ρ_{zu}. If zero lies outside this set, the implied orthogonality $E[z'u] = 0$ fails and the instrument is invalid.

To operationalize this insight, we propose the Correlation Restriction (CR) test, a frequentist procedure that integrates partial identification methods with union-bound inference.[1] Drawing on the partial identification literature (Tamer, 2010; Stoye, 2009; Imbens and Manski, 2004), we construct confidence intervals (CIs) for each bound and aggregate them using union-bound methods to achieve valid coverage. Following the intersection-union principle (Chernozhukov et al., 2013; Casella and Berger, 2021), as applied by Conley et al. (2012); Kolesár and Rothe (2018); Ban and Kedagni (2022), this approach trades conservatism for guaranteed coverage.

Recent work by Kaido et al. (2019) shows that, when constraints are strongly informative, alternative bounding procedures can deliver tighter confidence intervals than plug-in projection methods. Building on this idea, Bei (2024) proposes refined union bound approaches designed to enhance statistical power. We construct CIs for the identified set of ρ_{zu} using a modified conditional union-bound (MCUB) approach (Bei, 2024), which controls overall test size, ensures uniform coverage for the joint inequalities, and improves power relative to standard Bonferroni corrections.

The CR test then evaluates whether the null value $\theta_0 \equiv (\rho_{zu} = 0)$ lies within this confidence inter-

[1] For more details on union bounds, see Pishro-Nik (2014); Lehmann and Romano (2022) (ch. 9.3) present the Bonferroni correction as a direct application.

val. If $0 \notin$ CI, the data reject instrument validity at level α. Intuitively, the test quantifies how far an instrument can deviate from perfect exogeneity before its correlation with u becomes statistically incompatible with the model's correlation restrictions. Theory or prior empirical evidence on the sign of $\rho_{xu} \equiv \operatorname{corr}(x,u)$ provides the structural constraint required for this assessment. Restricting the admissible range $\mathscr{D}$ to either positive or negative values increases the discriminatory power of the test relative to procedures that assume $\rho_{xu} \in (-1,1)$. Simulations confirm that the CR test reliably distinguishes valid from invalid instruments. Instruments that fail the correlation-restriction criterion correspond to known sources of endogeneity bias, while genuinely exogenous instruments are never falsely rejected. In canonical empirical settings such as Card (1993)'s returns-to-education model, analyses of criminal recidivism, and development policy studies, the test identifies the subset of instruments consistent with statistical exogeneity and provides a transparent diagnostic of IV credibility.

Our contribution is threefold. First, we extend DiTraglia and García-Jimeno (2021)'s correlation-restriction framework to develop a test for instrument validity when the sign of $\operatorname{corr}(x,u)$ is known. Second, we provide a practical implementation using the delta method and Bei (2024)'s modified conditional union-bound approach, yielding finite-sample confidence intervals. Third, we demonstrate that sign restrictions on endogeneity, as suggested by economic theory, enable testability in continuous IV models, bridging the divide between theoretical impossibility (Pearl, 1995; Gunsilius, 2021) and their use in practice.

In this context, our innovation, relative to existing research, is methodological. We recognize that theoretically motivated sign restrictions allow for testing the validity of instruments within the framework of partial identification. We have developed a formal procedure to carry out this test. While we build on the correlation bounds established by Di Traglia (2021) and the inference methods proposed by Bei (2024), our contribution lies in the testing procedure itself. Specifically, we demonstrate that sign information enables a formal evaluation of the restriction $\rho_{zu} = 0$ using strict containment criteria.To assess the practical conservativeness of set coverage without compromising the clarity of the primary procedure, we also include an Imbens–Manski/Stoye parameter-coverage interval as a clearly labelled diagnostic.

This study contributes to the evolving literature on testing identifying assumptions in econometric models. Pearl (1995) introduced instrumental inequalities, a theoretical constraint for instrument validity that required discretization of the endogenous treatment, while conjecturing that validity testing becomes infeasible with continuous treatments. Subsequent research has extended and refined these ideas along multiple trajectories. Bonet (2001) formally proved Pearl's conjecture for discrete outcomes, while Manski (2003) established an equivalent inequality in the context of missing data models. Kitagawa (2015) developed a formal test for continuous outcomes with binary treatments and instruments, and Wang et al. (2017) contributed empirical testing procedures specifically for binary data structures. Kédagni and Mourifié (2020) comprehensively confirmed Pearl's conjecture for binary variables, generalizing the results to accommodate discrete instruments with arbitrary outcome distributions. Jiang and Ding (2020) derived identification bounds for binary models that incorporate measurement error, and Acerenza et al. (2023) applied a similar methodology to test identifying assumptions within bivariate probit specifications. In parallel, a strand of research has emerged that assesses identifying assumptions by directly testing exclusion restrictions, without relying on conventional orthogonality conditions (Kiviet, 2020; D'Haultfœuille et al., 2024).

Despite important theoretical developments, applied econometrics still lacks fully developed methods for testing instrument validity in models with continuous endogenous variables. A notable exception is Gunsilius (2021), who demonstrated that imposing weaker structural assumptions can restore partial testability. More recent developments stem from machine-learning approaches (Xie et al., 2022, 2020; Silva and Shimizu, 2017), which, while innovative, offer limited practical guidance for applied econometric analysis. These approaches establish IV validity based on the causal structure of the model (G) and the independent noise terms (IN). A variable is a valid instrument if it meets the Graphical Independent Noise (GIN) condition, which is represented through a directed acyclic graph (DAG) that shows the dependencies among variables.The Instrumental Variable–Generalised Independent Noise (IV-GIN) method constructs a surrogate outcome variable representing the part of Y unexplained by treatment X and instrument Z. In non-Gaussian settings, a valid instrument Z is statistically independent of this surrogate, whereas an invalid one is not, providing a data-driven test for instrument validity.

Our paper addresses this critical gap by developing a rigorous test for instrument validity in continuous variable settings, grounded in partial identification methods and traditional orthogonality conditions. The methodological innovation lies in applying partial identification techniques (Tamer, 2010; Stoye, 2009; Imbens and Manski, 2004; Bei, 2024) to exploit the joint correlation restriction connecting instrument invalidity, treatment endogeneity, and the structural error term, as established by DiTraglia and García-Jimeno (2021), together with plausible sign restrictions on the endogeneity correlation $\mathrm{corr}(x,u)$. This approach produces a practical testing procedure that can be readily implemented using standard estimation methods.

The remainder of the paper is structured as follows. Section 2 presents the structural model and establishes the correlation restriction linking instrument validity to observable correlations. Section 3 develops the CR test, including the theoretical framework (Sections 3.3–3.4), the data-generating process for simulation validation (Section 3.6), comprehensive Monte Carlo evidence on the test's finite-sample performance (Section 3.7), and empirical applications (Sections 3.8–3.9). Section 4 concludes the study. Technical proofs and implementation details appear in the Appendices.

2 Model

We begin with the standard linear triangular system:

$$y = \beta x + u, \tag{1}$$

$$x = \pi z + \varepsilon. \tag{2}$$

where y is the outcome, x the endogenous regressor, z a candidate instrument, and u, ε unobservables. If additional exogenous controls exist, the model can be written in reduced form after projecting them out (see Appendix A.1).

2.1 Assumptions and Correlation Restriction

The analysis is based on three key assumptions.

Assumption 2.1 (Assumption 1 (Triangular model and relevance)) *The researcher observes i.i.d. draws of (y,x,z) from a linear triangular system with finite second moments. The instrument is relevant, $\mathrm{cov}(z,x) \neq 0$, and the regressor is endogenous, $\mathrm{cov}(x,u) \neq 0$. Instrument validity, $\rho_{zu} = 0$, is not assumed and is the object of inference. Only a valid instrument satisfies $E[z'u] = 0$.*

Assumption 2.2 (Assumption 2 (Covariance structure)) *The covariance matrix of (u,ε,z), denoted Ω, exists and is positive definite.*

We denote the vector of observable correlations by $\rho \equiv (\rho_{xy}, \rho_{xz}, \rho_{yz})'$. Under Assumptions 1–2, DiTraglia and García-Jimeno (2021) show that

$$\rho_{zu} = \rho_{xz} r_{xu} - (\rho_{xy}\rho_{xz} - \rho_{yz}) \sqrt{\frac{1 - r_{xu}^2}{1 - \rho_{xy}^2}}, \tag{3}$$

where r_{xu} denotes a candidate value of the unobserved correlation ρ_{xu}. This relationship links regressor endogeneity (ρ_{xu}) and instrument validity ($\rho_{zu} = 0$) through observable correlations, providing the theoretical foundation for our test.

Assumption 2.3 (Assumption 3 (Asymptotics for sample correlations)) $\sqrt{n}\,(\hat{\rho} - \rho) \xrightarrow{d} N(0, \Sigma)$ *for some positive definite* Σ*, and* $\hat{\Sigma}$ *is a consistent estimator of* Σ *(e.g., multivariate normality of* (x, y, z) *is sufficient).*

Equation (3) defines a mapping $\rho_{zu} = g(\rho, r_{xu})$ from the unobserved correlation ρ_{xu} to the implied ρ_{zu}. The function $\hat{\rho}_{zu} = g(\hat{\rho}, r_{xu})$ with $\hat{\rho} = (\hat{\rho}_{xz}, \hat{\rho}_{xy}, \hat{\rho}_{yz})'$ is twice continuously differentiable in a neighborhood of ρ. Given that ρ_{xu} is not observed, the model determines only the *identified set*

$$\mathscr{C} = \{\, g(\rho, r_{xu}) : r_{xu} \in \mathscr{D} \,\},$$

where $\mathscr{D} \subset (-1, 1)$ is the admissible range of endogeneity correlations consistent with a positive-semidefinite correlation matrix. The bounds of this set are

$$\underline{C} = \min_{r_{xu} \in \mathscr{D}} \left[\rho_{xz} r_{xu} - (\rho_{xy}\rho_{xz} - \rho_{yz}) \sqrt{\tfrac{1 - r_{xu}^2}{1 - \rho_{xy}^2}} \right], \tag{4}$$

$$\overline{C} = \max_{r_{xu} \in \mathscr{D}} \left[\rho_{xz} r_{xu} - (\rho_{xy}\rho_{xz} - \rho_{yz}) \sqrt{\tfrac{1 - r_{xu}^2}{1 - \rho_{xy}^2}} \right]. \tag{5}$$

Assumption 2.4 (Assumption 4 (Compactness of the restriction set)) *The admissible set* $\mathscr{D} \subseteq (-1, 1)$ *is nonempty and compact (i.e., closed and bounded).*

Lemma 2.1 *Consider the linear triangular system* (1)–(2) *under Assumptions 1–4. The estimator of* ρ_{zu} *defined by the mapping* $g(\rho, r_{xu})$ *in equation* (3) *is consistent but biased.*

Proof: See Appendix A.2. The proof uses Assumptions 1–2 to ensure the correlation restriction (3) holds, and Assumption 3 to establish the asymptotic properties of the sample correlations.

2.2 Identified-Set Framework and Instrument Validity

As Stoye (2009) emphasizes, there is an important distinction between constructing confidence intervals that cover the identified set versus those that cover the parameter of interest. Set coverage requires $\Pr(\mathscr{C} \subseteq \mathrm{CI}) \geq 1-\alpha$, while parameter coverage requires only $\inf_{\theta_0 \in \mathscr{C}} \Pr(\theta_0 \in \mathrm{CI}) \geq 1-\alpha$. The implication

$$\mathscr{C} \subseteq \mathrm{CI} \;\Rightarrow\; \theta_0 \in \mathrm{CI} \quad \forall \theta_0 \in \mathscr{C}$$

shows that set coverage implies parameter coverage, but not vice versa. Consequently, confidence intervals designed for set coverage are generally wider—and hence more conservative—than those designed only for parameter coverage.

In our context, the parameter of primary interest is $\theta_0 \equiv \rho_{zu} = 0$, representing instrument validity. We adopt set coverage via the MCUB method as our main inferential approach, which automatically satisfies parameter coverage but may be overly conservative for testing $\theta_0 = 0$. To assess the practical impact of this conservativeness, we additionally report results from the Imbens–Manski/Stoye parameter-coverage interval $\mathrm{CI}^{IM}_{1-\alpha}$ as a diagnostic. This dual approach allows us to evaluate whether the conservativeness of set coverage materially affects our conclusions.

Our simulations indicate that this conservatism is not a serious practical concern: only instruments in close proximity to the valid instrument, with small deviations from orthogonality with the error term, pass the CR test (see Table 1). Moreover, the set-coverage approach offers several advantages: (i) it is more conservative, reducing the risk of falsely rejecting valid instruments; (ii) it provides a complete characterization of the identified set, which is informative beyond the binary validity decision; and (iii) the MCUB method offers computational tractability with well-understood theoretical properties.

In our setting, $\theta_0 \equiv \mathrm{corr}(z,u) = 0$ is the unique condition under which the IV estimator consistently identifies the structural parameter β. Any deviation from this condition, $\mathrm{corr}(z,u) \neq 0$, implies that β_{IV} is biased. In the linear model $y = x\beta + u$ with $E(u) = 0$, multiplying both sides by z and taking expectations yields

$$E(zy) = E(zx)\beta + E(zu) = E(zx)\beta + \theta_0,$$

where $\theta_0 \equiv E(zu)$. The exogeneity condition for a valid IV requires $\theta_0 = 0$. With centered variables (or equivalently, zero-mean errors and demeaned instruments), this implies

$$E(zu) = 0 \Leftrightarrow \text{Cov}(z,u) = 0 \Leftrightarrow \rho_{zu} = 0.$$

This covariance-based formulation is more transparent, requires fewer implicit assumptions, and connects directly to our correlation-based framework.

Accordingly, we treat the correlation parameter $\rho_{zu} \equiv \text{corr}(z,u)$ as the central measure of IV validity. This insight provides the foundation for constructing the *correlation-restriction (CR) test*, which formally evaluates whether the observed data are compatible with instrument exogeneity under the identified-set framework developed above.

2.2.1 Properties of the True Identified Set $\mathscr{C}$ for a Valid Instrument

Since the identified set $\mathscr{C}$ forms the foundation of the correlation-restriction test, it is essential to establish its basic properties and the restrictions that can be imposed on its domain.

Lemma 2.2 *Consider the linear triangular system* (1)–(2). *Let Assumptions 1–4 hold and let* $\rho = (\rho_{xy}, \rho_{xz}, \rho_{yz})'$. *For fixed* ρ *with* $|\rho_{xy}| < 1$, *the function* $r_{xu} \mapsto g(\rho, r_{xu})$ *defined in* (3) *is continuous on* $\mathscr{D}$. *Consequently, if* $r_{xu} \in \mathscr{D}$ *almost surely, then* $\rho_{zu} = g(\rho, r_{xu}) \in \mathscr{C}$ *almost surely, where* $\mathscr{C} = \{g(\rho, r) : r \in \mathscr{D}\}$.

Proof See Appendix A.3. The proof uses Assumption 2 (positive definiteness of Ω), and verifies continuity of g in r_{xu} given $|\rho_{xy}| < 1$ and $r_{xu} \in \mathscr{D} \subseteq (-1,1)$ (Assumption 4).

Since $g(\rho, \cdot)$ is continuous and $\mathscr{D}$ is compact, the image $\mathscr{C} = \{g(\rho, r) : r \in \mathscr{D}\}$ is a compact interval, i.e., $\mathscr{C} = [\underline{C}, \overline{C}]$ with $\underline{C} = \min_{r \in \mathscr{D}} g(\rho, r)$ and $\overline{C} = \max_{r \in \mathscr{D}} g(\rho, r)$.

Knowledge of the sign of $\text{corr}(x,u)$ narrows the feasible domain:

$$\mathscr{D} = \begin{cases} (0,1), & \text{if } \text{corr}(x,u) > 0, \\ (-1,0), & \text{if } \text{corr}(x,u) < 0. \end{cases}$$

Restricting $\mathscr{D}$ in this way improves power by avoiding overly wide identified sets. However, a researcher can conduct a sensitivity analysis using a subset of $\mathscr{D}$ and ascertain over what range of assumed $\text{corr}(x,u)$

values the instrumental variable may not be rejected as invalid.

In addition, instrument validity requires that $\rho_{zu} = 0$ holds. Therefore, when using the function $g(\rho, r_{xu})$ defined above, we should ensure that $0 \in \mathscr{C}$. To do so, we introduce one additional assumption, which does not conflict with the overall logic of the proposed test. Specifically, in the following lemma, we assume that $\rho_{xz} > 0$, a condition that simplifies the proof. This assumption is made without loss of generality, since if $E(z'u) = 0$ holds, then for $z_1 = -z$, the orthogonality condition $E(z_1'u) = 0$ also holds.

Lemma 2.3 *Consider the linear triangular system* (1)–(2) *under Assumptions 1–4. Let* $\rho_{xz} > 0$ *and define*

$$g(r_{xu}) = \rho_{xz} r_{xu} - h\sqrt{\frac{1 - r_{xu}^2}{1 - \rho_{xy}^2}},$$

where $h = \rho_{xy}\rho_{xz} - \rho_{yz}$. *Then:*

1. *if* $\mathscr{D} \subset (0,1)$, *a unique* $r_{xu}^* \in \mathscr{D}$ *satisfies* $g(r_{xu}^*) = 0$ *if and only if* $h > 0$*;*
2. *if* $\mathscr{D} \subset (-1,0)$, *a unique solution exists if and only if* $h < 0$.

Proof See Appendix A.4. The proof uses Assumption 2 to ensure that $|\rho_{xy}| < 1$, so the function g is well-defined, and applies monotonicity arguments to establish uniqueness.

Lemma 2.3 implies that for a valid instrument ($\rho_{zu} = 0$), the admissible domain $\mathscr{D}$ must contain a unique r_{xu}^* consistent with the observed correlations.

2.3 Asymptotic Behavior of Sample Correlations

For any pair (x, y), the sample correlation is

$$\hat{\rho}_{xy} = \frac{s_{xy}}{s_x s_y},$$

and it satisfies the asymptotic distribution

$$\sqrt{n}(\hat{\rho}_{xy} - \rho_{xy}) \xrightarrow{d} N(0, (1 - \rho_{xy}^2)^2).$$

Fisher's transformation

$$\hat{\rho}'_{xy} = \frac{1}{2}\log\left(\frac{1 + \hat{\rho}_{xy}}{1 - \hat{\rho}_{xy}}\right)$$

yields an asymptotic variance of one. By the continuous mapping theorem and the Glivenko–Cantelli theorem, $\hat{\rho}_{xy} \xrightarrow{p} \rho_{xy}$, so that

$$\hat{\rho} = (\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz})' \xrightarrow{d} \mathscr{N}(\mu, \Sigma).$$

To demonstrate the consistency of an estimator of the identified set, given that $\hat{\rho}_{xy}$, $\hat{\rho}_{xz}$, $\hat{\rho}_{yz}$ are consistent estimators, we must show that the estimated set converges to the true identified set in a precise sense (Tamer, 2010; Stoye, 2009). For instance, one can establish consistency by showing that the Hausdorff distance between the estimated and true sets converges to zero as the sample size $n \to \infty$. Alternatively, one can assess consistency by evaluating the coverage probability of the true identified set by the estimated confidence interval.

Lemma 2.4 (Consistency of the estimated identified set) *Consider the linear triangular system* (1)–(2) *under Assumptions 1–4. Let* $\hat{\mathscr{C}}_n = [\hat{\underline{C}}, \hat{\overline{C}}]$ *denote the estimated identified set constructed from* $\hat{\rho}$ *and* $\mathscr{D}$. *Under standard regularity conditions,*

$$\hat{\mathscr{C}}_n \xrightarrow{d_H} \mathscr{C},$$

where d_H *denotes the Hausdorff distance. Hence, the estimated set is consistent.*

Proof See Appendix A.5.

The proof uses Assumption 3 to obtain $\hat{\rho} \xrightarrow{p} \rho$ and assumes $\mathscr{D}$ (Assumption 4) is compact and g is continuous (Lemma 2.2), which together imply $\sup_{r \in \mathscr{D}} |g(\hat{\rho}, r) - g(\rho, r)| \xrightarrow{p} 0$ and hence convergence of the min/max endpoints.

These findings lay the probabilistic groundwork for constructing confidence intervals and conducting coverage tests, which are essential for developing the comprehensive inference procedure discussed in Section 3.

3 Inference and the Correlation-Restriction Test

Let $\theta \equiv \rho_{zu}$ denote the instrument–error correlation. Because the nuisance correlation ρ_{xu} is not observed, we impose a researcher-specified admissible set

$$\rho_{xu} \in \mathscr{D} \subset (-1, 1),$$

based on maintained sign restrictions and substantive prior information. For each $r \in \mathscr{D}$, the model implies

$$\theta(r) \equiv \rho_{zu}(r) = g(\rho, r), \qquad \rho \equiv (\rho_{xy}, \rho_{xz}, \rho_{yz})'.$$

In implementation we evaluate r on a finite grid $B = \{r_{xu,b} : b \in B\}$ and write

$$\lambda_b \equiv g(\rho, r_{xu,b}), \qquad \hat{\lambda}_b \equiv g(\hat{\rho}, r_{xu,b}).$$

Since each $r_{xu,b}$ implies a single value, the bounds at grid point b are degenerate: $\lambda_{\ell,b} = \lambda_{u,b} = \lambda_b$.

Identified set. The (grid-based) identified set for θ is the union-bound interval

$$\mathscr{C} = [\theta_\ell, \theta_u] \equiv \left[\min_{b \in B} \lambda_b,\ \max_{b \in B} \lambda_b\right]. \tag{6}$$

Instrument validity is the membership null

$$H_0 :\ 0 \in \mathscr{C} \qquad \text{vs.} \qquad H_1 :\ 0 \notin \mathscr{C}, \tag{7}$$

equivalently

$$\min_{b \in B} \lambda_b \le 0 \le \max_{b \in B} \lambda_b.$$

3.1 Identified-Set Estimation

Given the sample correlations $\hat{\rho} = (\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz})'$, the plug-in estimate of ρ_{zu} as a function of r_{xu} is

$$\hat{\rho}_{zu}(r_{xu}) = \hat{\rho}_{xz} r_{xu} - (\hat{\rho}_{xy}\hat{\rho}_{xz} - \hat{\rho}_{yz})\sqrt{\frac{1 - r_{xu}^2}{1 - \hat{\rho}_{xy}^2}}, \tag{8}$$

where $r_{xu} \in \mathscr{D} = [r_{\min}, r_{\max}]$ represents the researcher's prior beliefs about the correlation between x and u. The sample identified set is therefore

$$\hat{\mathscr{C}}_n = [\underline{\hat{C}}, \overline{\hat{C}}] = \left[\inf_{r_{xu} \in \mathscr{D}} \hat{\rho}_{zu}(r_{xu}),\ \sup_{r_{xu} \in \mathscr{D}} \hat{\rho}_{zu}(r_{xu})\right],$$

with bounds obtained by minimizing and maximizing equation (8) over $\mathscr{D}$.

3.2 Asymptotic Distribution via the Delta Method

Let $g(\rho, r_{xu})$ denote the population mapping in equation (3) and write $\hat{g} = g(\hat{\rho}, r_{xu})$. The gradient of g with respect to ρ is

$$\nabla_\rho g(\rho, r_{xu}) = \begin{pmatrix} -\rho_{xz}\sqrt{S} + (\rho_{xy}\rho_{xz} - \rho_{yz})\frac{S\rho_{xy}}{1-\rho_{xy}^2} \\ r_{xu} - \rho_{xy}\sqrt{S} \\ \sqrt{S} \end{pmatrix}, \tag{9}$$

where $S = (1 - r_{xu}^2)/(1 - \rho_{xy}^2)$.

Under standard regularity conditions (e.g., multivariate normality of (x, y, z) suffices),

$$\sqrt{n}\,(\hat{\rho} - \rho) \xrightarrow{d} N(0, \Sigma),$$

where Σ is the asymptotic covariance matrix of $\hat{\rho} = (\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz})$. We estimate Σ using the nonparametric bootstrap: on each of B resamples drawn with replacement from the data, we recompute the vector of sample correlations, and $\hat{\Sigma}$ is taken as the sample covariance matrix of these bootstrap replications,

$$\hat{\Sigma} = \frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\rho}^{(b)} - \bar{\rho}\right)\left(\hat{\rho}^{(b)} - \bar{\rho}\right)^\top, \tag{10}$$

where $\hat{\rho}^{(b)}$ is the vector of sample correlations in the b-th resample and $\bar{\rho} = B^{-1}\sum_{b=1}^{B}\hat{\rho}^{(b)}$ is their mean. Under standard regularity conditions this bootstrap estimator is consistent for the asymptotic covariance matrix of the sample correlations (see, e.g., van der Vaart, 1998). This full covariance structure is particularly important when correlations are moderate to large, as the off-diagonal elements capture joint sampling variation that would be ignored by a diagonal approximation.

Applying the multivariate delta method, for each fixed $r_{xu} \in \mathscr{D}$,

$$\sqrt{n}\big(g(\hat{\rho}, r_{xu}) - g(\rho, r_{xu})\big) \xrightarrow{d} N(0, V_g(r_{xu})), \qquad V_g(r_{xu}) = \nabla_\rho g(\rho, r_{xu})'\Sigma\nabla_\rho g(\rho, r_{xu}).$$

Let $\widehat{V}_g(r_{xu})$ denote the plug-in estimate obtained by substituting $(\hat{\rho}, \hat{\Sigma})$.

Although this pointwise delta-method approximation yields the usual marginal interval $\mathrm{CI}_\Delta(r_{xu})$, our inference target is the *identified set* $\mathscr{C} = [g_{\min}(\rho), g_{\max}(\rho)]$ and valid inference requires *simultaneous* control of the inequalities defining $\mathscr{C}$. Accordingly, we do not base testing on pointwise intervals;

instead, we use the modified conditional union-bound method of Bei (2024) (Section 3.3) to construct $\mathrm{CI}_{\mathrm{MCUB}}$ for $\mathscr{C}$ and to implement the membership-based validity test.

3.3 Union-Bound Inference for Identified Sets

The modified conditional union-bound (MCUB) approach of Bei (2024) constructs a sharper confidence region for a scalar parameter θ that is partially identified by two moment inequalities:

$$m_\ell(\theta,\rho) = \theta - g_{\min}(\rho) \geq 0, \qquad m_u(\theta,\rho) = g_{\max}(\rho) - \theta \geq 0,$$

where $g_{\min}(\rho) = \min_{r_{xu}} g(\rho; r_{xu})$ and $g_{\max}(\rho) = \max_{r_{xu}} g(\rho; r_{xu})$. Linearizing $g(\rho; r_{xu})$ around the sample estimate $\hat{\rho}$ gives

$$g(\rho; r_{xu}) \approx g(\hat{\rho}; r_{xu}) + \nabla_\rho g(\hat{\rho}; r_{xu})'(\rho - \hat{\rho}).$$

This expansion converts the bounding functions into approximate moment inequalities for θ. In implementation we evaluate $g(\hat{\rho}; r_{xu})$ on a fine grid over $\mathscr{D}$, so the procedure applies to a finite collection of bounds as in Bei (2024).

MCUB construction. Let $\hat{g}_{\min} \equiv g_{\min}(\hat{\rho})$ and $\hat{g}_{\max} \equiv g_{\max}(\hat{\rho})$, where

$$g_{\min}(\hat{\rho}) = \min_{r_{xu} \in \mathscr{D}} g(\hat{\rho}; r_{xu}), \qquad g_{\max}(\hat{\rho}) = \max_{r_{xu} \in \mathscr{D}} g(\hat{\rho}; r_{xu}).$$

The sample moment functions are then

$$\hat{m}_\ell(\theta) = \theta - \hat{g}_{\min}, \qquad \hat{m}_u(\theta) = \hat{g}_{\max} - \theta.$$

Let $(\hat{\sigma}_\ell(\theta), \hat{\sigma}_u(\theta))$ be corresponding standard error estimates computed from the linearization and $\widehat{\mathrm{Var}}(\hat{\rho})$. Under H_0 both inequalities must hold, i.e., $\hat{m}_\ell(\theta) \geq 0$ and $\hat{m}_u(\theta) \geq 0$ up to sampling error. We therefore studentize the *violations* of these constraints:

$$\frac{-\hat{m}_\ell(\theta)}{\hat{\sigma}_\ell(\theta)} = \frac{\hat{g}_{\min} - \theta}{\hat{\sigma}_\ell(\theta)}, \qquad \frac{-\hat{m}_u(\theta)}{\hat{\sigma}_u(\theta)} = \frac{\theta - \hat{g}_{\max}}{\hat{\sigma}_u(\theta)},$$

which are positive only when the corresponding inequality is violated. Accordingly, we define the studentized test statistic as the following:

$$T_n(\theta) = \max\left\{\frac{-\hat{m}_\ell(\theta)}{\hat{\sigma}_\ell(\theta)}, \frac{-\hat{m}_u(\theta)}{\hat{\sigma}_u(\theta)}\right\}.$$

Following Bei (2024), we compute a conditional critical value $\hat{c}_c(\theta;\alpha_c)$ from a conditional pivot that exploits the dependence between the two studentized inequalities. We then form the *modified* conditional critical value by truncating $\hat{c}_c(\theta;\alpha_c)$ from below at a data-dependent truncation level $\hat{c}_t(\theta)$:

$$\hat{c}_m(\theta;\alpha) = \max\{\hat{c}_c(\theta;\alpha_c), \hat{c}_t(\theta)\}.$$

Here $\alpha_c \in (\alpha/2, \alpha)$ is a tuning parameter (we follow Bei (2024) and set $\alpha_c = 0.8\alpha$). This modification delivers uniform size control while typically yielding a sharper confidence region, particularly when the two bounds are close or coincide.

Confidence region by inversion. The MCUB confidence region is obtained by test inversion:

$$\mathrm{CI}_{\mathrm{MCUB}} = \{\theta \in \mathbb{R} : \; T_n(\theta) \le \hat{c}_m(\theta;\alpha)\}. \tag{11}$$

Since the identified set is an interval $\mathscr{C} = [\underline{C}, \overline{C}]$ and the inequalities are monotone in θ, the inverted acceptance set is an interval $[\underline{L}, \overline{L}]$. By construction,

$$\Pr\big(\mathscr{C} \subseteq \mathrm{CI}_{\mathrm{MCUB}}\big) \ge 1 - \alpha.$$

Based on the above discussion we state the following definition:

Definition 1 $H_0 : 0 \in \mathscr{C}$ (instrument validity is consistent with the identified set).

$H_1 : 0 \notin \mathscr{C}$ (instrument validity is inconsistent with the identified set).

This is consistent with eq.(7) and with the rejection rule. Under instrument validity, $\rho_{zu} = 0$, which necessarily implies $0 \in \mathscr{C}$; hence rejection of H_0 constitutes evidence against instrument validity.

The following discussion establishes that the CR test controls size at level α.

Membership test of instrument validity. Instrument validity is assessed by testing whether the null value $\theta \equiv \rho_{zu} = 0$ lies within this region:

$$\text{Reject } H_0 \quad \text{if} \quad 0 \notin \mathrm{CI}_{\mathrm{MCUB}}. \tag{12}$$

Equivalently, the p-value is computed by inversion as

$$p = \inf\{\alpha \in (0,1) : \ 0 \notin \mathrm{CI}_{\mathrm{MCUB},\,1-\alpha}\}.$$

If $p < \alpha$, the data are statistically inconsistent with instrument exogeneity.

Under H_0, the rejection event satisfies

$$\{0 \notin \mathrm{CI}_{\mathrm{MCUB}}\} \ \subseteq \ \{\mathscr{C} \nsubseteq \mathrm{CI}_{\mathrm{MCUB}}\},$$

since $0 \in \mathscr{C}$ and $\mathscr{C} \subseteq \mathrm{CI}_{\mathrm{MCUB}}$ jointly imply $0 \in \mathrm{CI}_{\mathrm{MCUB}}$. Therefore,

$$\Pr(\text{reject } H_0) \ = \ \Pr(0 \notin \mathrm{CI}_{\mathrm{MCUB}}) \ \leq \ \Pr\big(\mathscr{C} \nsubseteq \mathrm{CI}_{\mathrm{MCUB}}\big) \ \leq \ \alpha,$$

so the test controls size at level α.

Clarification of notation. To avoid confusion, we distinguish between two related but different objects:

- The **estimated identified set** $\hat{\mathscr{C}}_n = [\underline{\hat{C}}, \overline{\hat{C}}]$ is a *point estimate* of the true identified set $\mathscr{C} = [\underline{C}, \overline{C}]$. It is computed by substituting the sample correlations $\hat{\rho}$ into the mapping $g(\cdot)$ and optimizing over $r_{xu} \in \mathscr{D}$.

- The **confidence interval** $\mathrm{CI}_{\mathrm{MCUB}}$ is a *random interval* that covers the true identified set $\mathscr{C}$ with probability at least $1-\alpha$. It is constructed from $\hat{\mathscr{C}}_n$ to account for sampling uncertainty, as detailed above.

The population identified set $\mathscr{C} = [\underline{C}, \overline{C}]$ is a function of the population moments ρ and is therefore non-random. By contrast, the plug-in estimate $\hat{\mathscr{C}}_n = [\underline{\hat{C}}, \overline{\hat{C}}]$ and the MCUB confidence region $\mathrm{CI}_{\mathrm{MCUB}}$ are random because they depend on $\hat{\rho}$. The MCUB region is obtained by inverting the test based on the studentized inequalities and the modified conditional critical value, using $\widehat{\mathrm{Var}}(\hat{\rho})$ to account for sampling uncertainty. Consequently, the coverage statement $\Pr(\mathscr{C} \subseteq \mathrm{CI}_{\mathrm{MCUB}}) \geq 1-\alpha$ concerns the true identified set $\mathscr{C}$, not the estimator $\hat{\mathscr{C}}_n$ (which is itself random). In typical samples, $\mathrm{CI}_{\mathrm{MCUB}}$ can be viewed as an uncertainty-adjusted enlargement of $\hat{\mathscr{C}}_n$, constructed to deliver simultaneous coverage of $\mathscr{C}$ at level $1-\alpha$.

3.4 Diagnostic Inference: Imbens–Manski/Stoye Parameter-Coverage Interval

To assess sensitivity to the conservativeness inherent in set-covering confidence regions, we additionally report a less conservative diagnostic based on Imbens and Manski (2004) and Stoye (2009), which targets coverage of the *parameter* θ (for $\theta \in \mathscr{C}$) rather than set inclusion.

Endpoint estimators. We estimate the endpoints of (6) by

$$\hat{\theta}_\ell := \min_{b \in B} \hat{\lambda}_b, \qquad \hat{\theta}_u := \max_{b \in B} \hat{\lambda}_b, \qquad \hat{\Delta} := \hat{\theta}_u - \hat{\theta}_\ell. \tag{13}$$

Let $\hat{b}_\ell \in \arg\min_{b \in B} \hat{\lambda}_b$ and $\hat{b}_u \in \arg\max_{b \in B} \hat{\lambda}_b$ denote the selected grid indices.

Standard errors for the endpoints. Suppose that $\hat{\rho}$ satisfies the usual asymptotic linearity condition,

$$\sqrt{n}(\hat{\rho} - \rho) \;\Rightarrow\; \mathscr{N}(0, \Sigma_\rho), \tag{14}$$

and that the minimizer and maximizer are asymptotically stable. By the delta method applied at the selected indices, a convenient plug-in variance estimator is

$$\hat{\sigma}_\ell^2 := \nabla_\rho g(\hat{\rho}, r_{xu,\hat{b}_\ell})^\top \hat{\Sigma}_\rho \nabla_\rho g(\hat{\rho}, r_{xu,\hat{b}_\ell}), \qquad \hat{\sigma}_u^2 := \nabla_\rho g(\hat{\rho}, r_{xu,\hat{b}_u})^\top \hat{\Sigma}_\rho \nabla_\rho g(\hat{\rho}, r_{xu,\hat{b}_u}), \tag{15}$$

so that $\widehat{\mathrm{se}}(\hat{\theta}_\ell) = \hat{\sigma}_\ell / \sqrt{n}$ and $\widehat{\mathrm{se}}(\hat{\theta}_u) = \hat{\sigma}_u / \sqrt{n}$.[2]

Imbens–Manski/Stoye critical value and confidence interval. Following Imbens and Manski (2004) and Stoye (2009), we calibrate the critical value to deliver coverage for the *parameter* θ uniformly over

[2] When the argmin/argmax is not unique or stability is a concern, a bootstrap for $(\hat{\theta}_\ell, \hat{\theta}_u)$ can be used instead of (15).

$\theta \in \mathscr{C}$, rather than set inclusion $\mathscr{C} \subseteq \text{CI}$. Define

$$\hat{\sigma}_{\max} := \max\{\hat{\sigma}_\ell, \hat{\sigma}_u\}.$$

Let c_α^{IM} be the unique solution in $c \geq 0$ to

$$\Phi\left(c_\alpha^{IM} + \frac{\sqrt{n}\,\hat{\Delta}}{\hat{\sigma}_{\max}}\right) - \Phi\big(-c_\alpha^{IM}\big) \;=\; 1 - \alpha, \tag{16}$$

where $\Phi(\cdot)$ is the standard normal cdf. When $\hat{\Delta} = 0$, (16) reduces to $c_\alpha^{IM} = z_{1-\alpha/2}$. We then construct the confidence interval

$$\text{CI}_{1-\alpha}^{IM} := \left[\hat{\theta}_\ell - \frac{c_\alpha^{IM}\,\hat{\sigma}_\ell}{\sqrt{n}},\;\; \hat{\theta}_u + \frac{c_\alpha^{IM}\,\hat{\sigma}_u}{\sqrt{n}}\right]. \tag{17}$$

Under (14) and standard regularity conditions for endpoint selection, $\text{CI}_{1-\alpha}^{IM}$ attains the intended asymptotic *parameter coverage* for any $\theta \in \mathscr{C}$, and is typically less conservative than procedures calibrated for strong set coverage.[3]

Membership test of instrument validity. We implement the associated level-α membership test by inverting (17):

$$\text{Reject } H_0 \quad \Longleftrightarrow \quad 0 \notin \text{CI}_{1-\alpha}^{IM} \quad \Longleftrightarrow \quad \hat{\theta}_\ell - \frac{c_\alpha^{IM}\,\hat{\sigma}_\ell}{\sqrt{n}} > 0 \;\text{ or }\; \hat{\theta}_u + \frac{c_\alpha^{IM}\,\hat{\sigma}_u}{\sqrt{n}} < 0. \tag{18}$$

Because $\text{CI}_{1-\alpha}^{IM}$ targets parameter coverage in the sense of Imbens and Manski (2004)–Stoye (2009), the rule (18) controls rejection probability under H_0 asymptotically, while avoiding the additional conservativeness inherent in confidence regions designed to satisfy $\Pr(\mathscr{C} \subseteq \text{CI}) \geq 1 - \alpha$. We emphasize that (18) is reported as a diagnostic, while our main conclusions are based on the MCUB procedure and its strong guarantee of set coverage.

3.5 Interpretation

The correlation-restriction (CR) test evaluates whether the joint correlation structure among (x, y, z) admits any $\rho_{xu} \in \mathscr{D}$ that makes $\rho_{zu} = 0$ consistent with equation (8). The test's discriminatory power

[3] In particular, $\text{CI}_{1-\alpha}^{IM}$ need not contain the entire identified set $\mathscr{C}$; this trade-off reduces conservativeness relative to set-covering confidence regions.

derives from two sources: (i) the partial identification framework, which constructs an identified set $\mathscr{C}$ of potential instrument–error correlations compatible with the observed data, and (ii) sign restrictions on $\text{corr}(x,u)$, which narrow the admissible domain $\mathscr{D}$ and tighten the identified set relative to procedures that assume $r_{xu} \in (-1,1)$.

When the confidence interval CI_{MCUB} does not contain zero, the data provide statistical evidence against the null hypothesis $\rho_{zu} = 0$. The strength of this evidence depends on the p-value: values near zero indicate decisive rejection, while intermediate values ($0.05 < p < 0.10$) suggest borderline significance requiring caution. Our primary conclusions are based on the MCUB procedure, which guarantees strong set coverage $\Pr(\mathscr{C} \subseteq \text{CI}_{\text{MCUB}}) \geq 1-\alpha$. For robustness, we additionally report results from the Imbens–Manski/Stoye parameter-coverage interval $\text{CI}^{IM}_{1-\alpha}$ (Section 3.4), which may offer higher power under the maintained asymptotic approximation but does not guarantee set inclusion. When the two procedures yield different conclusions, the MCUB result should take precedence, though substantial disagreement may signal borderline validity worthy of further investigation.

Several practical considerations guide implementation. First, correct specification of $\text{sign}[\text{corr}(x,u)]$ is essential: misspecifying the direction of endogeneity can lead to spurious rejections or failures to detect invalid instruments. Researchers should ground this assumption in economic theory, institutional knowledge, or previous empirical evidence about the likely direction of omitted-variable bias, selection effects, or measurement error. Second, the test evaluates instrument validity conditional on the maintained assumptions (linearity, correct functional form, no omitted interaction terms), so rejection could reflect either instrument invalidity or model misspecification. Careful specification testing should precede the application of the CR test. Third, when multiple instruments are available, the test should be applied to each candidate separately, allowing researchers to rank instruments by p-value and select the most credible one for estimation.

Power enhancement through magnitude restrictions and sensitivity analysis. A limitation of using only sign restrictions on ρ_{xu} is that confidence intervals for θ_0 often contain zero, reducing the test's power to reject invalid instruments. This occurs because the domain $\mathscr{D} = (0,1)$ (or $(-1,0)$) remains wide, yielding a broad identified set $\mathscr{C}$ that typically includes zero.

Power can be enhanced by imposing *magnitude* restrictions on r_{xu} in addition to sign restrictions. If researchers have prior information suggesting that endogeneity is not only positive but also substantial—for example, $r_{xu} \in (0.2, 0.8)$ rather than $(0, 0.8)$—the identified set narrows correspondingly, potentially excluding zero and leading to rejection of invalid instruments.

In this light, we propose a *sensitivity analysis* approach that reports the "valid range" $[\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}]$: the range of r_{xu} values at which the CR test would *not* reject instrument validity. Formally, when the sign of endogeneity is known to be positive (i.e., $\rho_{xu} > 0$), define

$$\underline{\rho}^*_{xu} = \inf\left\{\rho^* > 0 : 0 \in \mathrm{CI}_{\mathrm{MCUB}} \text{ when } \mathscr{D} = (\rho^*, \bar{\rho})\right\},$$

$$\overline{\rho}^*_{xu} = \sup\left\{\rho^* > 0 : 0 \in \mathrm{CI}_{\mathrm{MCUB}} \text{ when } \mathscr{D} = (0, \rho^*)\right\},$$

where $\bar{\rho}$ is the upper bound of the feasible domain (e.g., 0.8). The valid range $[\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}]$ identifies the set of ρ_{xu} restrictions under which the instrument is *accepted* as valid. The complement of this range constitutes the *breakdown range* $\mathscr{K} = (0, \bar{\rho}) \setminus [\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}]$, which represents the set of ρ_{xu} values for which the instrument is rejected. For instance, if $\underline{\rho}^*_{xu} = 0.3$ and $\overline{\rho}^*_{xu} = 0.7$, the instrument passes the test when researchers believe $0.3 \leq \rho_{xu} \leq 0.7$, but is rejected for $\rho_{xu} \in (0, 0.3) \cup (0.7, \bar{\rho})$.

These valid and breakdown ranges serve two purposes:

1. **Transparency**: They quantify how strong the magnitude restriction on endogeneity must be to reject the instrument, allowing readers to judge whether the required assumptions are plausible. A narrow valid range indicates that acceptance is sensitive to the specific magnitude of ρ_{xu} assumed, while a wide range suggests robust validity across many plausible values.

2. **Robustness check**: If the valid range is narrow and centered at high values (e.g., $[\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}] = [0.6, 0.7]$), the instrument passes only under strong endogeneity assumptions that may be difficult to justify. Conversely, if the valid range is wide (e.g., $[0.1, 0.75]$), the instrument's validity is robust to a broad range of plausible endogeneity magnitudes. If the breakdown range includes very small values (e.g., $\rho_{xu} < 0.05$), the instrument is rejected even under mild endogeneity assumptions.

In practice, researchers can compute $\mathrm{CI}_{\mathrm{MCUB}}$ for a sequence of restricted domains $\mathscr{D}_k = (\rho_k, \bar{\rho})$ and $\mathscr{D}'_k = (0, \rho_k)$ with $\rho_k \in \{0.1, 0.2, 0.3, \ldots\}$, and identify the values at which $0 \in \mathrm{CI}_{\mathrm{MCUB}}$. Specifically:

- $\underline{\rho}^*_{xu}$ is the smallest ρ_k such that restricting $\mathscr{D} = (\rho_k, \bar{\rho})$ leads to *acceptance*.

- $\overline{\rho}^*_{xu}$ is the largest ρ_k such that restricting $\mathscr{D} = (0, \rho_k)$ leads to *acceptance*.

This sensitivity analysis provides a more nuanced assessment of instrument validity than a single pass/fail criterion, revealing the robustness of the validity conclusion to different prior beliefs about the magnitude of endogeneity. In some cases, the valid range may be one-sided (e.g., if acceptance occurs only for $\rho_{xu} > \underline{\rho}^*_{xu}$ but not for any upper bound restriction), which provides information about the direction of sensitivity.

Further sensitivity analysis can be conducted using the Imbens–Manski/Stoye parameter-coverage diagnostics by modifying domain restrictions to assess $\Pr(0 \in \mathrm{CI}^{IM})$. These diagnostics can be used to test the robustness of the main inference based on MCUB.

Computational Implementation The bounds $\underline{\hat{C}}$ and $\overline{\hat{C}}$ can be computed efficiently by evaluating (8) over a fine grid of r_{xu} values in $\mathscr{D}$. In our implementation, we use 80 equally spaced points. The function $\hat{\rho}_{zu}(r_{xu})$ is typically unimodal, so standard optimization routines can also be employed. Further details on the computation procedure are provided in Appendix B.1.

We illustrate the performance of the CR test using simulated data, where the true data-generating process is fully known. This simulation framework allows us to assess the test's ability to correctly identify valid instruments, particularly when the sign of $\mathrm{corr}(x,u)$ is known in advance.

3.6 Data Generating Process

To generate the artificial dataset, we follow the approach developed by Jones and Pewsey (2009). They introduce a transformation function $H(x;\zeta,\kappa) = \sinh[\kappa \sinh^{-1}(x) - \zeta]$, $\zeta \in \mathbb{R}$ and $\kappa \in \mathbb{R}_+$. When applied to the normal cumulative distribution function (CDF), Skew-Normal $:= S(x;\zeta,\kappa) = \Phi[H(x;\zeta,\kappa)]$, this transformation produces a unimodal distribution whose parameters ζ and κ control skewness and kurtosis, respectively Notably, setting $\zeta = 0$ and $\kappa = 1$ recovers the original normal distribution.

Generating the endogenous regressor. First, we generate sequence ν of N numbers.Using this sequence, we construct the endogenous variable $x = 5 \cdot H(\nu, 0, 0.9) + \text{Skew-Normal}(0, 1, 1.2)$, where $H(\cdot)$ is the Jones and Pewsey (2009) transformation. Next, we generate the disturbance term u, which is correlated with x.

Construction of the error term u and its correlation with x. We generate the structural error $\mathbf{u}$ with two components:

1. **Endogenous component** (correlated with **x**):

$$e = \lambda(x - \bar{x}) + \varepsilon, \quad \varepsilon \sim N(0, \sigma_x^2),$$

where $\lambda > 0$ is a scaling parameter that controls the degree of endogeneity, $\bar{x}$ is the sample mean of x, and ε is an independent noise term drawn from a normal distribution. We assume $\lambda > 0$, so the term $\lambda(x - \bar{x})$ induces positive correlation between u and x, while ε adds exogenous variation.

2. **Exogenous component** (uncorrelated with x):

$$u_1 = \text{Uniform}(-1, 1) + \text{Skew-Normal}(0, 1, 1.2),$$

where we extract the residual v from regressing u_1 on x, then rescale:

$$v = v \cdot \frac{\bar{x}}{2} \frac{sd(x)}{sd(v)}.$$

Sign of endogeneity. The total structural error is constructed as $u = e + v$. Since $\lambda > 0$ by assumption and v is orthogonal to x, this construction ensures $\text{cov}(x, u) > 0$, making the sign of endogeneity known and positive.

Generating the outcome variable. Then, we compute the outcome variable as follows:

$$y = 2x + u. \tag{19}$$

This structural equation implies that the true causal effect is $\beta = 2$.

Finally, we construct instrumental variables: z_i, $i = 0, 1, ..., 10$, satisfying $corr(z, u) \in (-0.2, ..., 0, ...0.2)$. In all cases, the correlation between the instruments and the endogenous variable remains below 0.5, except for one case, reflecting realistic empirical settings.

In our simulations, we set λ such that the resulting correlation $\rho_{xu} \approx 0.3$, representing substantial but not extreme endogeneity—a realistic scenario in many empirical applications where omitted variable bias is present but not dominant.

Domain restriction $\mathscr{D}$ in the simulations. Since the DGP ensures $\text{corr}(x, u) > 0$ by construction, the admissible domain for the CR test is set to $\mathscr{D} = (0, 0.8)$, reflecting the maintained assumption that endogeneity is positive. The upper bound of 0.8 (rather than 1) avoids numerical instabilities near the boundary of the correlation space and reflects the practical consideration that extremely high endogeneity correlations are rare in empirical settings. This domain restriction is the key identifying assumption that enables the CR test: by ruling out negative values of ρ_{xu}, we narrow the identified set $\mathscr{C}$ and increase the test's power to detect invalid instruments.

We conduct additional sensitivity analysis using the Imbens–Manski/Stoye parameter-coverage diagnostics verifying $Pr(0 \in CI_{\mathrm{IM}})$ by varying the domain restrictions from $(0, 0.8)$ to a narrow set of $(0.4, 0.6)$. The findings of such diagnostics can serve as a robustness test for the main inference (MCUB).

This DGP allows us to evaluate the CR test under controlled conditions, in which the true extent of instrument invalidity is known. By generating instruments with varying correlations to the error term—from strongly negative to positive—we can assess the test's ability to identify valid instruments (where $r_{zu} = 0$) and to reject those that violate the exogeneity condition. Since the direction of endogeneity is established (with $\mathrm{corr}(x, u) > 0$), we can also evaluate how effectively the test uses this information to improve inference. The next subsection presents the results of this simulation study.

3.7 Test Results

In this section, we first present the results of the CR test from a simulation study, followed by the results of the CR test applied to a single representative dataset.

3.7.1 Simulation Design and Aggregate Performance

We evaluate the finite-sample performance of the CR test through a Monte Carlo simulation study consisting of 100 independent replications. Each replication generates a sample of 800 observations according to the data-generating process specified in equation (8). Within each sample, we construct eleven candidate instruments $(Z_i, i = 0, \ldots, 10)$ with systematically varying correlations to the structural error term, spanning the range $\mathrm{corr}(Z_i, u) \in (-0.32, 0.36)$. For each instrument in each replication, we compute the CR test statistics and determine whether the instrument passes or fails the validity criterion. This design allows us to assess both Type I error control and statistical power (ability to detect invalid instruments) across a spectrum of instrument quality.

Table 1 summarizes the results across all instruments. Several key metrics are reported: the mean sample correlation between each instrument and the endogenous regressor (r_{xz}), the estimate of the true population correlation with the error term (r_{zu}), the mean IV estimate (β_{IV}), and three probability measures central to the CR test. In all cases, the true structural parameter is $\beta = 2$.

Table 1: Simulation Results Across Instruments: 100 Replications

Metric	Z_0	Z_1	Z_2	Z_3	$\mathbf{Z_4}$	$\mathbf{Z_5}$	$\mathbf{Z_6}$	Z_7	Z_8	Z_9	Z_{10}
r_{xz}	0.420	0.471	0.488	0.501	0.513	0.521	0.531	0.534	0.540	0.546	0.549
$sd(r_{xz})$	0.022	0.022	0.023	0.023	0.023	0.024	0.024	0.024	0.024	0.024	0.024
r_{zu}	-0.320	-0.189	-0.133	-0.087	$\mathbf{-0.032}$	**0.002**	**0.057**	0.079	0.126	0.196	0.359
$sd(r_{zu})$	0.024	0.023	0.023	0.023	0.022	0.021	0.020	0.020	0.019	0.019	0.017
β_{IV}	-0.890	0.474	0.956	1.340	**1.760**	**2.015**	**2.400**	2.560	2.880	3.365	4.490
$sd(\beta_{IV})$	0.340	0.235	0.209	0.189	0.168	0.156	0.140	0.133	0.122	0.110	0.110
p_0	0.00	0.001	0.052	0.466	**0.976**	**1.000**	**1.000**	0.933	0.210	0.0005	0.000
Imbens–Manski/Stoye parameter-coverage diagnostics: $\Pr(0 \in \mathrm{CI}_{\mathrm{IM}})$											
$\mathscr{D} = [0.0, 0.8]$	0.65	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0
$\mathscr{D} = [0.4, 0.8]$	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	0.0	0.0
$\mathscr{D} = [0.4, 0.6]$	0.0	0.0	0.08	0.88	1.0	1.0	1.0	1.0	0.72	0.0	0.0

Notes: p_0 = proportion of replications in which $0 \in \mathrm{CI}_{\mathrm{MCUB}}$ on domain $\mathscr{D} = [0.0, 0.8]$. Bold entries highlight instruments near the validity boundary ($r_{zu} \approx 0$) for which $0 \in CI_{\mathrm{MCUB}}$ (fail to reject $H_0 : 0 \in \mathscr{C}$ at 5%); Z_5 represents the case where r_{zu} is approximately zero. $N = 800$ observations per replication. By construction, $\rho_{xu} > 0$. Imbens–Manski/Stoye parameter-coverage rates computed for three domain restrictions as indicated.

The statistic p_0, which measures whether zero is contained in the CI, exceeds the nominal level 0.95 only for instruments Z_4, Z_5, and Z_6 with estimated average correlation with the error term, $r_{z_j u} \in (-0.032, 0.057)$ for $j \in [4,5,6]$ correspondingly. This outcome strongly suggests that a test based solely on point inclusion exhibits discriminatory power, given the range $\mathscr{D}$ is correctly determined.

The proportion of replications with $0 \in CI_{MCUB}$ varies systematically based on the magnitude of the instrument-error correlation. Instruments with moderate deviations, like Z_3 (47% at $r_{zu} = -0.09$) and Z_7 (93% at $r_{zu} = 0.08$), exhibit reduced reliability. Instruments with large violations of orthogonality—Z_0 (at $r_{zu} = -0.32$) and Z_{10} (at $r_{zu} = 0.36$)—consistently fail across all 100 replications, whereas —Z_1 (at $r_{zu} = -0.19$) and Z_9 (at $r_{zu} = 0.196$)— fail at the 5% significance level. This trend confirms the CR test's effectiveness in distinguishing valid from invalid instruments based on correlation.

The additional Imbens–Manski/Stoye parameter-coverage diagnostics verifying $Pr(0 \in CI_{\mathrm{IM}})$ is overall in line with the main inference results. However, this diagnostic manifests greater sensitivity to the domain restriction (see Table 1). Only when the domain $\mathscr{D}$ is narrowly restricted around the true value of $\rho_{xu} \approx 0.5$ (which is known in the artificial DGP case), the inference results coincide with those based on the MCUB procedure.

3.7.2 Single-Sample Detailed Analysis

Having established the test's aggregate performance across repeated samples, we now examine its behavior in a single representative dataset. This analysis illustrates how applied researchers would implement the CR test with a fixed sample and provides detailed insight into the relationships among the various test statistics.

For this single-sample application of 2000 observations, we evaluate whether the point null value $\theta_0 = 0$ (representing $\rho_{zu} = 0$) lies within the CI_{MCUB}. Given that the data-generating process satisfies $\rho_{xu} > 0$ by construction, we evaluate the test over the restricted domain $r_{xu} \in [0, 0.8]$, consistent with the maintained assumption about the direction of endogeneity.

Tables 2 and 3 present the detailed results. Table 2 examines six instruments with non-negative instrument–error correlations ($r_{zu} \geq 0$), including the valid instrument Z_5 with $r_{zu} = 0$. Table 3 focuses on five instruments with negative correlations ($r_{zu} \leq 0$), again including Z_5 as a benchmark. Both tables confirm that the point null value $\rho_{zu} = 0$ lies within $\mathrm{CI}_{\mathrm{MCUB}}$ for instruments Z_i, $i \in [3, ..., 7]$, with the confidence regions at the 95% level.

The Imbens-Manski/Stoye supplementary diagnostics for parameter coverage, which check $Pr(0 \in CI_{\mathrm{IM}})$, indicate that when a broader domain $\mathscr{D} = [0.0, 0.8]$ is employed in the MCUB method, instrumental variables (IVs) with slight deviations from orthogonality ($r_{zu} \in (-0.32, 0.36)$) are not rejected, while others are rejected. Tables 2 and 3 illustrate that this domain restriction increases the sensitivity of the diagnostics. This heightened sensitivity occurs because more constrained domain assumptions—including the correct value of ρ_{xu}—enhance the inferences drawn from these diagnostics.

3.7.3 Interpretation and Practical Implications

The simulation evidence establishes several key findings. First, inference based on whether zero is contained in the confidence interval exhibits clear discriminatory power, reliably distinguishing valid from invalid instruments: rejections are consistent for $|r_{zu}| \geq 0.35$ and acceptance is consistent for instruments near perfect orthogonality. Second, the CR test identifies a subset of instruments $\{Z_4, Z_5, Z_6\}$, with $\rho_{zu} \in (-0.032, 0.057)$, as potentially valid. This subset contains the truly valid instrument Z_5,

Table 2: Single-Sample Test Results: Instruments with $r_{zu} \geq 0$

Metric	$\mathbf{Z_5}$	$\mathbf{Z_6}$	$\mathbf{Z_7}$	Z_8	Z_9	Z_{10}
ρ_{xz}	0.56	0.57	0.57	0.58	0.58	0.59
r_{zu}	**0.00**	**0.06**	**0.08**	0.13	0.20	0.27
β_{IV}	2.02	2.39	2.54	2.85	3.30	3.76
Plug-in Set	[-0.07, 0.08]	[-0.01, 0.13]	[0.01, 0.15]	[0.06,0.20]	[0.13, 0.26]	[0.21, 0.33]
$\mathrm{CI}_{\mathrm{MCUB}}$	[-0.11, 0.12]	[-0.05,0.17]	[-0.02, 0.18]	[0.03, 0.23]	[0.10, 0.30]	[0.18, 0.36]
CI (Simple)	[-0.40, 0.34]	[-0.34, 0.38]	[-0.31, 0.39]	[-0.26, 0.43]	[-0.18, 0.47]	[-0.09, 0.52]
$0 \in \mathrm{CI}_{\mathrm{MCUB}}$	✓	✓	✓	×	×	×
p (zero)	**1.00**	**1.00**	**1.00**	0.004	0.00	0.00
Additional Imbens–Manski/Stoye parameter-coverage diagnostics.						
$\mathscr{D} = [0.4, 0.6]$: CI_{IM}	[-0.11,0.12]	[-0.05,0.16]	[-0.027,0.18]	[0.028,0.23]	[0.10,0.29]	[0.18,0.36]
$0 \in CI_{\mathrm{IM}}$	✓	✓	✓	×	×	×
$\mathscr{D} = [0.2, 0.6]$: CI_{IM}	[-0.24,0.12]	[-0.18,0.17]	[-0.15,0.19]	[-0.10,0.23]	[-0.02,0.29]	[0.06,0.36]
$0 \in CI_{\mathrm{IM}}$	✓	✓	✓	✓	✓	×
$\mathscr{D} = [0.0, 0.8]$: CI_{IM}	[-0.35,0.29]	[-0.29,0.33]	[-0.27,0.35]	[-0.22,0.38]	[-0.14,0.43]	[0.014,0.52]*
$0 \in CI_{\mathrm{IM}}$	✓	✓	✓	✓	✓	×

Notes: Results for six instruments from a single sample of 2000 observations. The true DGP is $y = 2x + u$. By construction, $\rho_{xu} > 0$."MCUB" refers to the method of Bei (2024); "Simple" denotes the standard union-bound approach. ✓ indicates $0 \in CI_{\mathrm{MCUB}}$ (fail to reject $H_0 : 0 \in \mathscr{C}$ at 5%); × indicates rejection. p (zero) reports the corresponding p-value. Boldface denotes instruments that are not rejected by the CR test. The lower panel reports Imbens–Manski/Stoye parameter-coverage diagnostics under the domain restrictions shown; *marks the outside interval that excludes zero (for $r_{xu} = 0.33$).

Table 3: Single-Sample Test Results: Instruments with $r_{zu} \leq 0$

Metric	Z_0	Z_1	Z_2	$\mathbf{Z_3}$	$\mathbf{Z_4}$	$\mathbf{Z_5}$
ρ_{xz}	0.49	0.50	0.52	0.53	0.55	0.56
r_{zu}	-0.24	-0.19	-0.13	$-$**0.09**	$-$**0.03**	**0.00**
β_{IV}	0.11	0.56	1.02	1.38	1.78	2.02
Plug-in Set	[-0.32,-0.16]	[-0.27,-0.11]	[-0.21,-0.05]	[-0.16,-0.01]	[-0.11, 0.04]	[-0.07, 0.08]
$\mathrm{CI}_{\mathrm{MCUB}}$	[-0.36,-0.12]	[-0.31,-0.07]	[-0.25,-0.02]	[-0.20, 0.03]	[-0.14, 0.08]	[-0.11, 0.11]
CI (Simple)	[-0.67, 0.15]	[-0.62, 0.20]	[-0.56, 0.24]	[-0.51, 0.27]	[-0.44, 0.34]	[-0.46, 0.34]
$0 \in \mathrm{CI}_{\mathrm{MCUB}}$	×	×	×	✓	✓	✓
p (zero)	0.00	0.00	0.013	**1.00**	**1.00**	**1.00**
Additional Imbens–Manski/Stoye parameter-coverage diagnostics.						
$\mathscr{D} = [0.4, 0.6]$: CI_{IM}	[-0.35,-0.12]	[-0.30,-0.07]	[-0.24,-0.007]	[-0.19,0.03]	[-0.14,0.08]	[-0.10,0.11]
$0 \in CI_{\mathrm{IM}}$	×	×	×	✓	✓	✓
$\mathscr{D} = [0.2, 0.6]$: CI_{IM}	[-0.48,-0.12]	[-0.42,-0.07]	[-0.37,-0.02]	[-0.32,0.03]	[-0.27,0.08]	[-0.24,0.11]
$0 \in CI_{\mathrm{IM}}$	×	×	×	✓	✓	✓
$\mathscr{D} = [0.0, 0.8]$: CI_{IM}	[-0.69,-0.02]*	[-0.53,0.13]	[-0.49,0.18]	[-0.44,0.22]	[-0.39,0.26]	[-0.35,0.29]
$0 \in CI_{\mathrm{IM}}$	×	✓	✓	✓	✓	✓

Notes: Results for six instruments from a single sample of 2000 observations. The true DGP is $y = 2x + u$. By construction, $\rho_{xu} > 0$."MCUB" refers to the method of Bei (2024); "Simple" denotes the standard union-bound approach. ✓ indicates $0 \in CI_{\mathrm{MCUB}}$ (fail to reject $H_0 : 0 \in \mathscr{C}$ at 5%); × indicates rejection. p (zero) reports the corresponding p-value. Boldface denotes instruments that are not rejected by the CR test. The lower panel reports Imbens–Manski/Stoye parameter-coverage diagnostics under the domain restrictions shown; *marks the outside interval that excludes zero (for $r_{xu} = -0.35$).

demonstrating that the test effectively narrows the candidate set by screening on the correlation between the instrument and the error term. Third, instruments with $|r_{zu}| > 0.1$ are rejected at the α significance level, while those in the range $0.09 < |r_{zu}| < 0.13$ exhibit borderline behaviour, as expected near the boundary of the identified set. Taken together, these results indicate that zero-inclusion tests possess sufficient power to discriminate among instruments of varying quality without requiring strong parametric assumptions.

The Imbens–Manski/Stoye parameter-coverage diagnostics corroborate the $\mathrm{CI}_{\mathrm{MCUB}}$-based conclusions, but prove more sensitive to the choice of domain $\mathscr{D}$. Since the true range of ρ_{xu} is rarely known with precision in practice, the $\mathrm{CI}_{\mathrm{MCUB}}$ approach offers more robust performance across plausible domain specifications, which is a practical advantage offsetting its theoretical conservativeness.

3.8 Applying the IV correlation restrictions test to empirical data

We illustrate the practical utility of the CR test by revisiting Card's (1993) seminal study on the returns to education. Card estimated the following wage equation:

$$lwage_i = \beta \cdot educ_i + H_i' \cdot \alpha + u_i \tag{20}$$

where H_i' includes experience, race, location and regional dummies. Card (1993) employed four instrumental variables: proximity to two-year ($nearc2$) and four-year colleges ($nearc4$), as well as parental education levels ($fatheduc$, $motheduc$). In this setting, the primary source of endogeneity arises from omitted ability bias, which induces a positive correlation between education and the regression error term (i.e., $corr(educ, u) > 0$). Other potential sources of endogeneity include model misspecification and nonlinearities, particularly "sheepskin effects" associated with degree completion. The presence of positive correlation implies that OLS estimates are likely to overstate the true causal returns to education.

The analysis reveals that all IV estimates of returns to education exceed the OLS estimate. Using $nearc4$ as an instrument yields a return of 0.132, about 76% higher than the OLS estimate of 0.075, while $nearc2$ provides an even larger but statistically insignificant estimate of 0.293. Parental education instruments ($fatheduc$ and $motheduc$) yield statistically significant estimates that are 20-37% higher than OLS. These surprising results, where IV estimates surpass OLS despite expected positive ability bias, may be due to factors like downward bias in OLS from measurement error, heterogeneous returns,

Table 4: CR Test Results Across Z vectors: Card case

Metric	$fatheduc$	$motheduc$	$nearc2$	$nearc4$
Plug-in Set	[0.071, 0.841]	[0.138, 0.875]	[0.752, 1.000]	[0.285, 0.938]
CI_{MCUB}	[0.067, 0.843]	[0.130, 0.880]	[0.729, 1.009]	[0.269, 0.945]
CI (Simple)	[0.063, 0.845]	[0.123, 0.883]	[0.681, 1.000]	[0.255, 0.951]
$0 \in \text{CI}_{\text{MCUB}}$	$\times$	$\times$	$\times$	$\times$
p (zero)	0.00	0.00	0.00	0.00
Number of obs.	2320	2657	3010	3010
Additional Imbens–Manski/Stoye parameter-coverage diagnostics.				
CI_{IM}	[0.069, 0.842]	[0.134, 0.877]	[0.737, 1.000]	[0.276, 0.941]
$0 \in \text{CI}_{\text{IM}}$	$\times$	$\times$	$\times$	$\times$

Notes: This table reports inference results for four instruments. "MCUB" refers to the method of Bei (2024); "Simple" refers to the standard union-bound approach. The above results are determined for $r_{xu} \in (0, 0.8]$. ✓ indicates $0 \in \text{CI}_{\text{MCUB}}$ (fail to reject $H_0\colon 0 \in \mathscr{C}$ at 5%); $\times$ indicates rejection. p (zero) reports the corresponding p-value. The lower panel reports Imbens–Manski/Stoye parameter-coverage diagnostics under the domain restrictions shown. All four instruments are rejected across the full domain, consistent with the education–ability endogeneity prior ($\rho_{xu} > 0$).

and the potential invalidity of family background instruments. More recently, Chalak (2019) builds on Card (1995) to document a nonlinear return to education that is smaller than OLS estimates.

To implement the CR test, we compute correlations using reduced form variables: $y = (I - P_H)lwage$, $x = (I - P_H)educ$, and $z = (I - P_H)\tilde{z}$ for each instrument $\tilde{z}$, where P_H is the linear projection matrix onto H. The results in Table 4 indicate that all instruments are rejected as invalid at the 5% significance level over $r_{xu} \in (0, 0.8]$. The additional Imbens–Manski/Stoye parameter-coverage diagnostics support the results of the main CI_{MCUB}-based inference. The breakdown analysis confirms that no valid range exists within D = (0, 0.8): the instrument is rejected regardless of the assumed magnitude of endogeneity. This robust rejection aligns with the plausibility of a nonlinear return to education effects as highlighted by Chalak (2019).

3.9 Applications Across Diverse Research Settings

To evaluate the generalizability and empirical performance of the Correlation-Restriction (CR) test, we apply it to four influential instrumental-variable (IV) studies drawn from distinct research domains. Each application highlights how the test behaves under different forms of endogeneity and data structures.

3.9.1 Electronic Monitoring and Recidivism

Di Tella and Schargrodsky (2013) investigate the causal effect of electronic monitoring (EM) relative to imprisonment on criminal recidivism using the specification

$$R_i = \alpha + \beta \, EM_i + X_i'\gamma + \varepsilon_i, \tag{21}$$

where R_i indicates post-release detention, EM_i denotes assignment to electronic monitoring, and X_i includes controls for crime type, demographics, and judicial factors. Because judges tend to assign EM to lower-risk offenders, a positive selection bias is plausible. The authors instrument EM_i using (i) an indicator for whether the judge has previously assigned EM, and (ii) the judge's EM assignment rate excluding the current offender.

The CR test indicates that both instruments satisfy the validity conditions. For each instrument, zero is contained in CI_{MCUB}, so we fail to reject instrument validity at the 5% significance level (p = 1.00 for both instruments) (see Table 5, model: Di Tella and Schargrodsky, 2013). The additional Imbens–Manski/Stoye parameter-coverage diagnostics corroborate the results of the main CI_{MCUB}-based inference. The sensitivity analysis results indicate that the CR test would reject these IVs as invalid for $\rho_{xu} \in (0.25, 0.80)$; the instruments satisfy the CR test only within the valid range (0.0, 0.25).

3.9.2 School Access and Educational Outcomes

Burde and Linden (2013) examine the impact of school proximity on academic achievement in rural Afghanistan using

$$TestScore_i = \alpha + \beta \, Enrollment_i + H_i'\gamma + u_i, \tag{22}$$

where H_i includes demographic covariates. Endogeneity arises from unobserved parental characteristics influencing both enrollment and test performance ($\text{corr}(Enrollment, u) > 0$).

Using village school presence as an instrument, the CR test rejects validity across the entire range $r_{xu} \in (0, 0.8)$. The CR test strongly rejects the validity of this instrument, as zero is not contained in the confidence interval, so we reject instrument validity at the 5% significance level ($p = 0.0$) (see Table 5, model: Burde and Linden, 2013). The additional Imbens–Manski/Stoye parameter-coverage diagnostic corroborates the results of the main CI_{MCUB}-based inference. These results provide strong evidence against IV validity, consistent with the possibility that school location directly affects test outcomes through community resources, teacher quality, or school infrastructure.

The breakdown analysis confirms that no valid range exists within D = (0, 0.8): the instrument is rejected regardless of the assumed magnitude of endogeneity. This robust rejection aligns with the plausibility of direct school-infrastructure effects on test outcomes.

Table 5: CR Test Results Across Empirical Applications

	DS (2013)		BL (2013)	G. et al. (2011)	BI (2005)
Metric ↓| IVs→	Judge history	Judge preference	School access	Draft lottery	British conquest
Plug-in Set	[−0.244, 0.629]	[−0.044, 0.773]	[0.272, 0.933]	[−0.765, 0.011]	[−0.361, 0.530]
$\mathrm{CI}_{\mathrm{MCUB}}$	[−0.256, 0.638]	[−0.046, 0.775]	[0.253, 0.943]	[−0.769, 0.017]	[−0.375, 0.544]
CI (Simple)	[−0.267, 0.647]	[−0.048, 0.776]	[0.209, 0.965]	[−0.774, 0.024]	[−0.394, 0.563]
$0 \in \mathrm{CI}_{\mathrm{MCUB}}$	✓	✓	×	✓	✓
p (zero)	1.00	1.00	0.00	1.00	1.00
Number of obs.	1503	1503	687	5000	3541
Valid range	such that $\rho \in \mathscr{D}^* = (\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}) : 0 \in \mathrm{CI}_{\mathrm{MCUB}}(\mathscr{C})$				
$[\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}]$	[0.15, 0.25]	[0.0, 0.15]	NA	[−0.2, 0.0]	[−0.6, −0.5]
Additional Imbens–Manski/Stoye parameter-coverage diagnostics.					
$\mathrm{CI}_{\mathrm{IM}}$	[−0.254, 0.637]	[−0.045, 0.774]	[0.259, 0.938]	[−0.769, 0.017]	[−0.374, 0.542]
$0 \in \mathrm{CI}_{\mathrm{IM}}$	✓	✓	×	✓	✓

Notes: Results for five instruments. "MCUB" refers to the method of Bei (2024); "Simple" refers to the standard union-bound approach. $\mathrm{CI}_{\mathrm{IM}}$ is the Imbens–Manski/Stoye parameter-coverage interval. ✓ indicates $0 \in \mathrm{CI}_{\mathrm{MCUB}}$ (fail to reject $H_0: 0 \in \mathscr{C}$ at 5%); × indicates rejection. p (zero) reports the corresponding p-value. The lower panel reports Imbens–Manski/Stoye parameter-coverage diagnostics under the domain restrictions shown. NA indicates the instrument is rejected across the entire admissible domain.

Study abbreviations: DS (2013) = Di Tella & Schargrodsky (2013); BL (2013) = Burde & Linden (2013); G. et al. (2011) = Galiani et al. (2011); BI (2005) = Banerjee & Iyer (2005).

3.9.3 Military Service and Criminal Behavior

Galiani et al. (2011) estimate the causal effect of mandatory military service on subsequent criminal behavior using

$$CrimeRate_{ci} = \alpha + \beta\, Conscription_{ci} + \delta_c + u_{ci}, \tag{23}$$

where δ_c represents cohort fixed effects. Potential negative selection bias ($\mathrm{corr}(x,u) < 0$) may arise from systematic draft avoidance. To address this, the authors instrument conscription with draft-lottery eligibility.

The CR test strongly supports the validity of this instrument, as zero is contained in the confidence interval, so we fail to reject instrument validity at the 5% significance level ($p = 1.0$) for $r_{xu} \in [-0.8, 0]$ (see Table 5, model: Galiani et al., 2011). The diagnostic sensitivity analysis indicates that $[\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}] \in [-0.2, 0]$. This finding indicates that the IV used in Galiani et al. (2011) is consistent with a moderate negative correlation between the endogenous regressor and unobserved avoidance behavior. The additional Imbens–Manski/Stoye parameter-coverage diagnostic also fails to reject the validity of the instrument at the 5% significance level ($p = 1.0$), in line with the main CI_{MCUB}-based inference.

3.9.4 Colonial Institutions and Economic Development

Banerjee and Iyer (2005) study the long-term developmental effects of British colonial land-revenue systems in India. They show that regions where landlords held proprietary rights experienced lower agricultural investment, reduced productivity, and weaker social outcomes relative to regions where cultivators retained land rights. Specifically, they estimate

$$\text{PropHYadopt}_i = \alpha + \beta \, \text{NonLandlordShare}_i + V_i'\gamma + u_i, \tag{24}$$

where V_i is a vector of controls. Because British annexation targeted more productive districts, a downward OLS bias ($\text{corr}(x, u) < 0$) is likely. To address this, the authors instrument the non-landlord share with the timing of British conquest (1820–1856), during which administrative assignment of land systems was largely exogenous.

The CR test supports the validity of this instrument (see Table 5, model: Banerjee and Iyer, 2005), as zero is contained in the confidence interval, and we fail to reject instrument validity at the 5% significance level ($p = 1.00$) across the endogeneity range $r_{xu} \in (-0.8, 0)$. The diagnostic sensitivity analysis indicates that $[\underline{\rho}^*_{xu}, \overline{\rho}^*_{xu}] \in [-0.6, -0.50]$. This result aligns with the historical evidence that colonial expansion systematically targeted more productive territories, thus, the unobserved effect of this behavior may be quite strong as the sensitivity analysis indicates. The Imbens–Manski/Stoye parameter-coverage diagnostic supports the main inference based on CI_{MCUB}.

3.9.5 Synthesis of Empirical Evidence

Taken together, these empirical applications highlight the robustness and interpretive clarity of the CR test across domains ranging from criminology to education and economic history. By combining theoretical identification with feasible inference, the method effectively bridges the gap between structural econometric theory and empirical practice.

Several key patterns emerge. First, the CR test's conclusions reflect institutional reasoning about instrument credibility, with stronger rejections linked to questionable exclusion restrictions and instruments with strong institutional justification passing. Second, the test effectively addresses both positive and negative selection scenarios, allowing for discrimination based on the sign of the correlation. Third, p-values for the membership test vary significantly across applications (0.00 to 1.00), indicating that the test captures true variations in instrument quality rather than merely sample size or model specification differences. Fourth, the IM diagnostic corroborates the MCUB-based conclusion in all five applications examined, suggesting that the conservativeness of the set-covering approach does not materially affect empirical inference in these settings. Fifth, the breakdown analysis reveals a relatively narrow valid range for valid instruments: Banerjee and Iyer, 2005 (valid range [-0.6, -0.5]), Di Tella and Schargrodsky, 2013 (valid range [0.0–0.25]), and Galiani et al., 2011 (valid range [-0.2, 0.0]). In contrast, instruments that fail unconditionally, as discussed by Card, 1993 and Burde and Linden, 2013, do not have any valid range within the admissible domain. Understanding these valid ranges offers practitioners quantitative guidance for selecting instruments, moving beyond simple binary decisions on validity.

4 Conclusion

This paper develops and validates a novel *Correlation-Restriction* (CR) test for assessing instrumental-variable (IV) validity when the direction of endogeneity is known. By leveraging the essential orthogonality constraint that valid instruments must meet, the CR test offers applied researchers a practical, theoretically grounded tool for assessing instrument exogeneity under partial identification.

Building on the framework of DiTraglia and García-Jimeno (2021), we show that valid instruments must satisfy specific correlation restrictions among observable variables, and that violations of these restrictions signal potential instrument invalidity. Simulation evidence demonstrates that valid instruments consistently pass the CR test, whereas invalid instruments are reliably rejected, confirming the test's effectiveness as a diagnostic for IV credibility.

The practical utility of the CR test is demonstrated by our empirical applications, which include a re-evaluation of Card (1993)'s returns-to-education study. Each of the four instruments fails the test: there is no valid range for any of them, and they all contribute to estimation bias across the admissible domain. In light of the potential for nonlinear returns to education highlighted by **?**, this result raises doubts about their reliability and supports recent criticisms of traditional IV specifications in education research.

Across four additional applications in criminal justice, economic development, education, and military conscription, the CR test validates instruments in three cases and identifies invalidity in one. It also uncovers systematic heterogeneity in both instrument reliability and the magnitude of endogeneity across empirical contexts.

These results have broad implications for applied econometric work. When the direction of endogeneity is known, the CR test serves as a transparent diagnostic for instrument selection and reliability assessment. Future research can extend this framework by (i) endogenously determining the sign of $\mathrm{corr}(x,u)$, (ii) adapting the procedure to nonlinear or heterogeneous-effect models, and (iii) integrating the CR test with complementary IV validity diagnostics. Together, these extensions would further enhance the scope and robustness of empirical inference under partial identification.

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Data Availability Statement

1. **Card (1993).** Using Geographic Variation in College Proximity to Estimate the Return to Schooling. *NBER Working Papers*, No. 4483, National Bureau of Economic Research. Dataset: `wooldridge::card`
2. **Burde and Linden (2013).** Bringing Education to Afghan Girls: A Randomized Controlled Trial of Village-Based Schools. *American Economic Journal: Applied Economics*, 5, 27–40. Dataset: `http://doi.org/10.3886/E113861V1`
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Appendix

A Proofs

A.1 Reduced-Form Representation

When exogenous controls H are present in (1)–(2), the system can be written as

$$y = \beta x + H'\delta + \tilde{u}, \qquad x = \pi z + H'\gamma + \varepsilon.$$

The matrix H includes a column of ones and satisfies $E(H'\tilde{u}) = 0$ and HH' has a full column rank, $E(H'w)$ exists for $w = \{y, x, z\}$. Projecting y, x, and z on H yields residuals $\tilde{y} = M_H y$, $\tilde{x} = M_H x$, and $\tilde{z} = M_H z$ with $M_H = I - H(H'H)^{-1}H'$. The CR test then applies to $(\tilde{y}, \tilde{x}, \tilde{z})$ without loss of generality.

A.2 Proof of Lemma 2.1

Proof Recall the specific function for $\hat{\rho}_{zu}$. By Assumptions 1–2, the population restriction implies

$$\hat{\rho}_{zu} = g(\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz}, r_{xu}) = \hat{\rho}_{xz} r_{xu} - (\hat{\rho}_{xy}\hat{\rho}_{xz} - \hat{\rho}_{yz})\sqrt{\frac{1 - r_{xu}^2}{1 - \hat{\rho}_{xy}^2}}$$

with $r_{xu} \in \mathscr{D}$ fixed, and $\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz} \xrightarrow{p} \rho_{xy}, \rho_{xz}, \rho_{yz}$. We need to show that

$$\hat{\rho}_{zu} \xrightarrow{p} \rho_{zu} := g(\rho_{xy}, \rho_{xz}, \rho_{yz}, r_{xu})$$

and that any identified set defined via this function is consistently estimated.

Let us first show the consistency of inputs. For this, we assume: $\hat{\rho}_{xy} \xrightarrow{p} \rho_{xy} \in (-1,1)$; $\hat{\rho}_{xz} \xrightarrow{p} \rho_{xz} \in (-1,1)$; $\hat{\rho}_{yz} \xrightarrow{p} \rho_{yz} \in (-1,1)$. These assumptions are standard when dealing with sample correlations under i.i.d. data.

Next, we need to establish the continuity of the function g. We next verify that the mapping

$$g(\rho_{xy}, \rho_{xz}, \rho_{yz}, r_{xu}) = \rho_{xz} r_{xu} - (\rho_{xy}\rho_{xz} - \rho_{yz})\sqrt{\frac{1-r_{xu}^2}{1-\rho_{xy}^2}}$$

is continuous at all points in the domain $(-1,1)^3$, with $\rho_{xu} \in \mathscr{D}$ fixed. The function is composed of products, subtractions, and a square root over $(1-\rho_{xy}^2)$, which is well-defined and differentiable for $\rho_{xy} \in (-1,1)$. Since all components are smooth on this domain, the function g is continuous. So, the continuous mapping theorem applies to this case.

Now, we need to show the consistency of $\hat{\rho}_{uz}$ using the continuous mapping theorem. Since g is a continuous function, the following applies:

$$\hat{\rho}_{uz} = g(\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz}, r_{xu}) \xrightarrow{p} g(\rho_{xy}, \rho_{xz}, \rho_{yz}, r_{xu}) = \rho_{zu}$$

We can also obtain an explicit analytic expression for the leading $O(1/n)$ bias of

$$\hat{\rho}_{zu} = g(\hat{\rho}, r_{xu}), \qquad \hat{\rho} = (\hat{\rho}_{xz}, \hat{\rho}_{xy}, \hat{\rho}_{yz})',$$

treating $g(\cdot)$ as a twice-differentiable function of the four sample correlations. Since r_{xu} is fixed thus dropped in determining the bias. We can state the bias in the standard delta-method second-order form, then give closed-form entries of the Hessian $H_g(\rho)$. By Assumption 3, $\sqrt{n}(\hat{\rho}-\rho) \xrightarrow{d} N(0,\Omega)$. The $1/n$ bias of $g(\hat{\rho})$ is approximated by the second-order Taylor term:

$$E\big[g(\hat{\rho})\big] - g(\rho) \;\approx\; \frac{1}{2n}\sum_{i=1}^{3}\sum_{j=1}^{3}\frac{\partial^2 g}{\partial\rho_i\partial\rho_j}(\rho)\,\Omega_{ij} = \frac{1}{2n}\operatorname{trace}\!\big(H_g(\rho)\,\Omega\big)\,,$$

where $H_g(\rho)$ is the 3×3 Hessian matrix of second partial derivatives of g evaluated at the population correlations ρ, and Ω is the asymptotic covariance matrix of $\sqrt{n}\hat{\rho}$: $\Omega_{ij} = \mathrm{Cov}\left(\sqrt{n}(\hat{\rho}_i-\rho_i), \sqrt{n}(\hat{\rho}_j-\rho_j)\right)$. Therefore, we can state: $\hat{\rho}_{zu} \xrightarrow{p} g(\rho, r_{xu}) = \rho_{zu}$, but $E[\hat{\rho}_{zu}] \neq \rho_{zu}$ for finite n. That is, the estimator (3) is consistent but exhibits a finite-sample bias of order $1/n$.■

A.3 Proof of Lemma 2.2

Proof For fixed $\rho = (\rho_{xy}, \rho_{xz}, \rho_{yz})'$, write

$$g(\rho, r_{xu}) = \rho_{xz} r_{xu} - (\rho_{xy}\rho_{xz} - \rho_{yz})\sqrt{\frac{1-r_{xu}^2}{1-\rho_{xy}^2}}.$$

The map $r_{xu} \mapsto \rho_{xz} r_{xu}$ is continuous. Since $|\rho_{xy}| < 1$, the denominator $1-\rho_{xy}^2$ is strictly positive, and for $r_{xu} \in \mathscr{D} \subseteq (-1,1)$ the numerator $1-r_{xu}^2$ is nonnegative. Hence the square-root term is well-defined and continuous in r_{xu} (on $\mathscr{D}$). Therefore, $g(\rho, \cdot)$ is continuous on $\mathscr{D}$.

If $r_{xu} \in \mathscr{D}$ almost surely, then by definition of $\mathscr{C}$ as the image of $\mathscr{D}$ under $g(\rho, \cdot)$, we have $g(\rho, r_{xu}) \in \mathscr{C}$ almost surely.■

A.4 Proof of Lemma 2.3

Proof Write

$$g(r) = \rho_{xz} r - h\sqrt{\frac{1-r^2}{1-\rho_{xy}^2}}, \qquad h = \rho_{xy}\rho_{xz} - \rho_{yz},$$

and assume $|\rho_{xy}| < 1$ so the expression is well-defined for $r \in (-1,1)$. For $r \in (-1,1)$,

$$g'(r) = \rho_{xz} + h\frac{r}{\sqrt{1-\rho_{xy}^2}\sqrt{1-r^2}}.$$

(i) Suppose $\mathscr{D} \subset (0,1)$. If $h > 0$, then for all $r \in (0,1)$ the second term is positive, so $g'(r) > \rho_{xz} > 0$ and g is strictly increasing on $\mathscr{D}$. Moreover,

$$g(0) = -\frac{h}{\sqrt{1-\rho_{xy}^2}} < 0, \qquad g(1) = \rho_{xz} > 0,$$

so by continuity there exists a root in $(0,1)$, and strict monotonicity implies it is unique. Conversely, if $h \le 0$ then $g(0) \ge 0$ and $g(r) > 0$ for all $r > 0$ (since $\rho_{xz} r \ge 0$ and $-h\sqrt{\cdot} \ge 0$), hence no root exists on $(0,1)$.

(ii) Suppose $\mathscr{D} \subset (-1,0)$. If $h < 0$, then for all $r \in (-1,0)$ we have $hr > 0$, so again $g'(r) > \rho_{xz} > 0$

and g is strictly increasing on $\mathscr{D}$. Also,

$$g(-1) = -\rho_{xz} < 0, \qquad g(0) = -\frac{h}{\sqrt{1-\rho_{xy}^2}} > 0,$$

so a root exists in $(-1,0)$ by continuity and is unique by strict monotonicity. Conversely, if $h \geq 0$ then $g(r) < 0$ for all $r < 0$ (since $\rho_{xz} r < 0$ and $-h\sqrt{\cdot} \leq 0$), so no root exists on $(-1,0)$.■

A.5 Proof of Lemma 2.4

Proof By Assumption 3, let $\hat{\rho} \xrightarrow{p} \rho$ and define $\hat{g}(r) \equiv g(\hat{\rho}, r)$ and $g(r) \equiv g(\rho, r)$. Assume $\mathscr{D}$ is compact and $g(\rho, r)$ is jointly continuous in (ρ, r). By uniform continuity on compact sets,

$$\sup_{r\in\mathscr{D}} \left|\hat{g}(r) - g(r)\right| = \sup_{r\in\mathscr{D}} \left|g(\hat{\rho}, r) - g(\rho, r)\right| \xrightarrow{p} 0.$$

Therefore, by the continuous mapping (argmin/argmax) theorem,

$$\underline{\hat{C}} = \min_{r\in\mathscr{D}} \hat{g}(r) \xrightarrow{p} \min_{r\in\mathscr{D}} g(r) = \underline{C}, \qquad \overline{\hat{C}} = \max_{r\in\mathscr{D}} \hat{g}(r) \xrightarrow{p} \max_{r\in\mathscr{D}} g(r) = \overline{C}.$$

Convergence of the endpoints implies $\hat{\mathscr{C}_n} = [\underline{\hat{C}}, \overline{\hat{C}}] \xrightarrow{d_H} [\underline{C}, \overline{C}] = \mathscr{C}$, where d_H denotes the Hausdorff metric.■

B Additional discussions

B.1 Implementation Notes

The CR test can be implemented in any statistical software that supports matrix operations and re-sampling. A simple R-package that implements the CR test can be installed using

```
remotes::install_git("https://github.com/ratbekd/ivcrtest.git")
```

A minimal computational algorithm proceeds as follows:

Implementation Steps:

1. **Estimate sample correlations:** Compute $\hat{\rho} = (\hat{\rho}_{xy}, \hat{\rho}_{xz}, \hat{\rho}_{yz})'$ from the data.

2. **Evaluate on grid:** Evaluate $g(\hat{\rho}, r_{xu})$ over a fine grid $\{r_{xu,b} : b \in B\} \subset \mathscr{D}$ to obtain the sample identified set endpoints:

$$\hat{g}_{\min} = \min_{r_{xu} \in \mathscr{D}} g(\hat{\rho}; r_{xu}), \qquad \hat{g}_{\max} = \max_{r_{xu} \in \mathscr{D}} g(\hat{\rho}; r_{xu}).$$

3. **Estimate covariance matrix:** Estimate the asymptotic covariance matrix $\hat{\Sigma}$ of $\hat{\rho}$ using using the nonparametric bootstrap (see Section 3.2).

4. **Compute standard errors:** Obtain standard error estimates $\hat{\sigma}_\ell(\theta)$ and $\hat{\sigma}_u(\theta)$ for the endpoints using the delta method and linearization:

$$\hat{\sigma}_\ell^2 = \nabla_\rho g(\hat{\rho}, r_{xu,\hat{b}_\ell})^\top \hat{\Sigma} \nabla_\rho g(\hat{\rho}, r_{xu,\hat{b}_\ell}),$$

where $\hat{b}_\ell \in \arg\min_{b \in B} \hat{\lambda}_b$ (and similarly for $\hat{\sigma}_u$).

5. **Construct studentized test statistic:** Define the sample moment functions

$$\hat{m}_\ell(\theta) = \theta - \hat{g}_{\min}, \qquad \hat{m}_u(\theta) = \hat{g}_{\max} - \theta,$$

and compute the studentized test statistic:

$$T_n(\theta) = \max\left\{ \frac{-\hat{m}_\ell(\theta)}{\hat{\sigma}_\ell(\theta)}, \frac{-\hat{m}_u(\theta)}{\hat{\sigma}_u(\theta)} \right\}.$$

6. **Compute MCUB critical value:** Following Bei (2024), compute the conditional critical value $\hat{c}_c(\theta; \alpha_c)$ and the truncation level $\hat{c}_t(\theta)$, then form the modified conditional critical value:

$$\hat{c}_m(\theta; \alpha) = \max\{\hat{c}_c(\theta; \alpha_c), \hat{c}_t(\theta)\},$$

with tuning parameter $\alpha_c = 0.8\alpha$.

7. **Construct confidence region:** Obtain the MCUB confidence region by test inversion:

$$\mathrm{CI}_{\mathrm{MCUB}} = \{\theta \in \mathbb{R} : \ T_n(\theta) \le \hat{c}_m(\theta; \alpha)\}.$$

This region satisfies $\Pr(\mathscr{C} \subseteq \mathrm{CI}_{\mathrm{MCUB}}) \geq 1-\alpha$ by construction.

8. **Test instrument validity:** Reject $H_0 : 0 \in \mathscr{C}$ at level α if and only if $0 \notin \mathrm{CI}_{\mathrm{MCUB}}$. Alternatively, compute the p-value as

$$p = \inf\{\alpha \in (0,1) :\ 0 \notin \mathrm{CI}_{\mathrm{MCUB},\,1-\alpha}\}.$$

B.2 Extension to Overidentification and Multiple Endogenous Regressors

While the main analysis concentrates on regression models with a single endogenous regressor under exact identification, the framework can naturally be extended to more general settings.

B.2.1 Overidentified Models with a Single Endogenous Regressor

Consider an extension of the model specified in equations (1) and (2):

$$y = \beta x + v, \tag{A.1}$$

$$x = \pi_1 z_1 + \pi_2 z_2 + \cdots + \pi_n z_n + \varepsilon, \tag{A.2}$$

where we maintain a single endogenous regressor x but now incorporate n instrumental variables: $z_1, z_2, \ldots, z_n$.

In overidentified models, instrument validity must account for potential interdependence among instruments. We extend the Correlation Restriction (CR) test by requiring that each instrument individually satisfy the CR condition. For each instrument z_j, we test whether its correlation with the structural error term u is consistent with zero, given the assumed sign and magnitude of endogeneity $\mathrm{corr}(x,u)$. This ensures that all instruments are individually valid before being jointly employed in overidentified IV estimation.

B.2.2 Models with Multiple Endogenous Regressors

We now extend the framework to accommodate models with multiple endogenous regressors, where the number of instruments exceeds the number of endogenous variables:

$$\tilde{y} = \tilde{X}\beta + H'\gamma + v, \tag{A.3}$$

$$\tilde{X} = Z\Pi + \varepsilon, \tag{A.4}$$

where $\tilde{X}$ is an $n \times k$ matrix of endogenous regressors, H is a matrix of exogenous variables including a constant term, and $Z = [\tilde{z}_1, \tilde{z}_2, \ldots, \tilde{z}_n]$ is a matrix of n instrumental variables with $n > k$.

This setup allows for a generalized application of the Correlation Restriction (CR) test by assessing each instrument's validity individually for its corresponding endogenous regressor, while accounting for potential correlations among instruments and regressors. Such an extension provides a practical framework for evaluating instrument reliability in overidentified systems with multiple sources of endogeneity.

The key insight for extending our methodology to models with multiple endogenous regressors is to decompose the system into a series of single-endogenous-variable problems via appropriate projections. The procedure proceeds as follows:

1. **Instrument Assignment**: For each $\tilde{x}_i$ $(i = 1, 2, \ldots, k)$, we assign a specific subset of instruments $Z_i \subset Z$. This partitioning ensures that each endogenous regressor has a dedicated set of instruments, facilitating the identification of individual causal effects.

2. **First-Stage Projections**: For each endogenous variable $\tilde{x}_i$, we compute its projection onto its corresponding instrument space: $\hat{x}_i = P_{Z_i}\tilde{x}_i$, where P_{Z_i} denotes the linear projection matrix onto the space spanned by Z_i. This yields the matrix of fitted values $\hat{X}$. We additionally assume that each IV $z_i \in Z_i$ is uncorrelated with the first-stage errors of the other endogenous variables. That is, $E(z_i, \varepsilon_{-i}) = 0$.

3. **Sequential Residualization**: For each endogenous variable $\tilde{x}_i$, we define $H_i = [H|\hat{X}_{-\hat{x}_i}]$, where $\hat{X}_{-\hat{x}_i}$ is the matrix $\hat{X}$ excluding the column $\hat{x}_i$. We then compute residuals by projecting the relevant

variables onto the orthogonal complement of the space spanned by H_i:

$$z_j = (I - P_{H_i})\tilde{z}_j \quad \text{for each } z_j \in Z_i, \tag{A.5}$$

$$x_i = (I - P_{H_i})\tilde{x}_i, \tag{A.6}$$

$$y = (I - P_{H_i})\tilde{y}. \tag{A.7}$$

Because P_{H_i} includes fitted values of the other endogenous regressors, the residual v_i may contain first-stage errors whose correlation with Z_i is not zero in general. However, under the orthogonality assumption $E(z_i, \varepsilon_{-i}) = 0$, the resulting single-equation models satisfy the CR test requirements.

4. **Application of CR Test**: After residualization, the transformed variables follow:

$$y = \beta_i x_i + v_i, \tag{A.8}$$

$$x_i = \sum_{z_j \in Z_i} \pi_j z_j + \varepsilon_i. \tag{A.9}$$

This transformation effectively reduces the multiple endogenous regressor case to a series of single-endogenous-regressor models, allowing the CR test to be applied to each instrument $z_j \in Z_i$ individually.

The procedure is then repeated for each endogenous regressor $\tilde{x}_i$, enabling comprehensive validity assessment for all instruments across all endogenous variables. This approach provides a systematic framework for applying the CR test in more complex models while preserving the interpretability and theoretical foundations established in the main text.

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